I’ve always found curious that there are two base cases: it wasn’t possible to find a single base graph because these two are fundamentally anisomorphic, but at the same time it’s quite remarkable that any other non-planar graph can be reconducted just to these two.

Anyway, you might wonder, how do we know both that there is no way to draw these graphs without intersections, and that there isn’t any simpler graph that’s not planar?

Well, Kuratowski proved it, so we can trust his theorem.

But in case you still have doubts, you can also try to scramble the vertices in figure 15.7 and see if you can find a planar embedding: sit comfortably, because it might take a while, till you realize it’s not possible!

The other half of the claim, that there isn’t any smaller non-planar graph, is easier to show. Let’s focus on K_5 and first look at a graph with fewer vertices, in particular K_4, shown in figure 15.8.